The Robovac Problem

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Abstract

This paper introduces the "Robovac Problem," a novel reinforcement learning (RL) environment modeling the challenges of robotic vacuum cleaning with limited battery life and environmental uncertainty. We investigate the impact of training dirtiness levels and battery state granularity on agent performance and generalization. Experiments using tabular Q-learning demonstrate a critical relationship between training conditions and generalization ability, highlighting the importance of a well-designed state representation for robust and efficient cleaning strategies. For example, generalization peaks at a relatively narrow range of dirtiness levels (85-95%), and an increasing number of battery states consistently outperformed others despite the increase in state size. This setup brings 19% cleaning effectiveness to over 89% cleaning effectiveness.

1 Introduction

Reinforcement learning (RL) (Sutton and Barto, 2018) offers a powerful framework for developing intelligent agents capable of operating effectively in complex and uncertain environments. A significant challenge in RL, particularly relevant to robotics, lies in creating agents that can successfully balance exploration, planning, and efficient resource management. This paper introduces the "Robovac Problem," a novel RL environment inspired by the challenges faced by robotic vacuum cleaners. Unlike simpler cleaning scenarios, Robovac demands the agent to learn effective strategies for navigating an environment with uncertain cleanliness, limited battery power, and the need for a timely return to the charging station.

The Robovac Problem, a relaxation of the Traveling Salesman Problem Gavish and Graves (1978), introduces a unique set of constraints. While revisiting locations is permitted, cleaning a tile only once is enforced. This, combined with a limited battery and the uncertainty of whether a tile is already clean, creates a complex decision-making problem involving both exploration (locating dirty tiles) and efficient action execution (cleaning and navigating). This setup models real-world challenges faced by resource-constrained robotic agents.

This research investigates the impact of two key factors on agent performance and generalization: (1) the training environment's dirtiness level, and (2) the granularity of the agent's battery state representation. We hypothesize that a well-designed state representation, combined with a carefully selected training regime, is critical for developing a robust

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and generalizable cleaning strategy. We utilize tabular Q-learning Watkins (1989) as our foundation, allowing for a clear and interpretable benchmark. Our experiments, detailed in Section 2, systematically evaluate the influence of these factors. The results, presented in Section 3, demonstrate a complex interplay between training conditions, state-space design, and generalization performance. Section 4 then analyzes the implications of these findings, and Section 5 summarizes our contributions and suggests avenues for future research.

2 Experiment

This section details the experimental setup and procedures used to evaluate the performance of the proposed reinforcement learning (RL) agent in the Robovac environment. We focused on two key aspects: the impact of the training environment's dirtiness level on the agent's generalization performance, and the influence of the number of discretized battery states on the agent's ability to learn effective cleaning and return-to-base strategies.

2.1 Environment Setup

Our experiments utilized a static 8×8 grid environment, bounded by impassable walls. While the grid itself was empty (no inherent obstacles), this setup allowed us to easily control the density of dirty tiles, representing different cleaning challenges. The charging station was randomly placed adjacent to a wall in each episode, using a consistent random seed across all trials to maintain controlled conditions while ensuring variation across episodes. The dirtiness level, defined as the probability of a tile being initially dirty, was varied systematically across experiments.

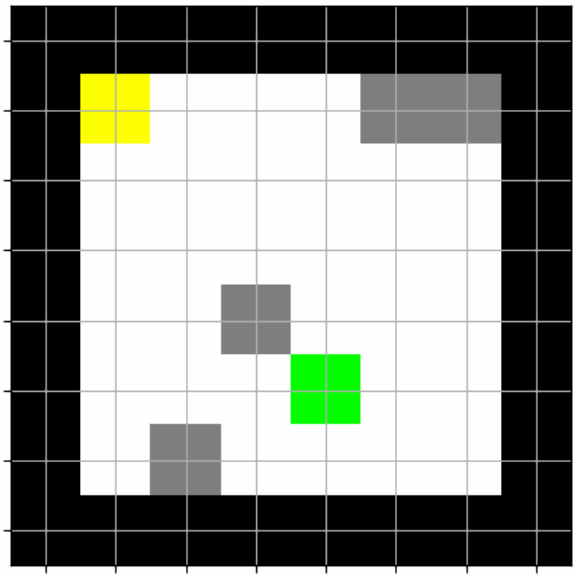


Figure 1: A representation of a Robovac environment. The black tiles are non-traversable, the white tiles are clean and the grey are dirty. The charging station is in the top left and the agent is the green tile in the lower center.

2.2 Agent and Algorithm

We employed a tabular Q-learning agent to solve the Robovac problem. Tabular Q-learning was chosen for its simplicity and interpretability, providing a clear benchmark for future work exploring more complex function approximation methods. The agent's actions included movement in one of four directions (up, down, left, right) and activating/deactivating the vacuum cleaner. The state representation included:

- **Agent Location:** The (x, y) coordinates of the agent on the grid.
- **Surrounding Tile Cleanliness:** The cleanliness (clean, dirty, or impassable) of the four tiles immediately adjacent to the agent on cardinal directions.
- **Battery State:** The discretized battery level, ranging from fully charged to critically low. The number of discretized states was varied across experiments (details below).

The Q-learning algorithm parameters were:

- **Learning Rate (α)**: 0.1
- **Discount Factor (γ)**: 0.99
- **Exploration Strategy**: ϵ-greedy, with ϵ annealed linearly from 0.75 to 0.01 over the first half of the training episodes.

The reward function was designed to incentivize efficient cleaning and timely return to the charging station:

- **Cleaning a Dirty Tile:** +10 reward
- **Cleaning a Clean Tile:** -10 penalty (to discourage unnecessary cleaning)
- **Movement:** -1 penalty for each movement step to encourage efficient path planning. A penalty of -2 was applied for changes in direction, representing turning time.
- **Episode Termination (Battery Depletion):** A significant penalty (-10 per remaining dirty tile) was enforced if the battery was depleted before returning to the base.

2.3 Experimental Design

Our experiments consisted of two primary phases:

2.3.1 Impact of Training Dirtiness Level

Agents were trained for 50,000 episodes on environments with various static dirtiness probabilities (0% to 100% in 5% increments). After training, each agent's generalization performance was assessed at widely varying settings of dirtiness probabilities (0% to 100% in 1% increments). This design allowed us to evaluate the impact of the training dirtiness level on the agent's ability to adapt to different environments. 20 agents were trained and tested for each dirtiness level to ensure statistically sound results.

2.3.2 Impact of Battery State Granularity

To investigate the impact of battery state representation, we conducted experiments with 2, 3, 4, and 5 discretized battery states. For each configuration, 20 agents were trained for 50,000 episodes in an environment with a fixed 85% dirtiness probability. Generalization testing was performed at 20%, 50%, and 95% dirtiness probabilities, allowing for more detailed tracking of generalization and the role of battery state granularity.

2.4 Performance Metrics

Agent performance was evaluated using two primary metrics:

- **Cleaning Effectiveness:** The percentage of dirty tiles successfully cleaned at the end of an episode.
- **Total Reward:** The cumulative reward obtained across the entire episode, reflecting the balance between cleaning efficiency and resource management (battery).

2.5 Baselines

For comparison purposes, we included a random policy baseline that selects a direction of travel randomly at each step and vacuums with perfect accuracy. This provided a lower bound for performance comparison.

3 Results

This section presents the quantitative results of our experiments, examining the influence of training dirtiness levels and the number of discretized battery states on agent performance.

3.1 Impact of Training Dirtiness Levels on Generalization

To assess the effect of the training environment's dirtiness level on generalization, we trained agents on environments with static dirtiness levels ranging from 0% to 100% (in 1% increments). Each agent was trained for 50,000 episodes at its assigned dirtiness level. Generalization performance was then evaluated by testing each agent on a range of unseen environments with widely varying dirtiness probabilities. Results were averaged across 20 independent agent runs for each training dirtiness level.

Figure 2 illustrates the cleaning effectiveness (percentage of dirty tiles cleaned) across different test environments, categorized by the dirtiness level used during training. Error bars represent the standard deviation across the 20 runs. Figure 3 displays the corresponding total reward obtained, reflecting the trade-off between cleaning efficiency and resource management. The figures clearly show a non-monotonic relationship between training dirtiness and generalization performance. Optimal generalization performance was achieved with training dirtiness levels within a narrow range, 85%–95%. Outside this optimal range, performance deteriorated significantly.

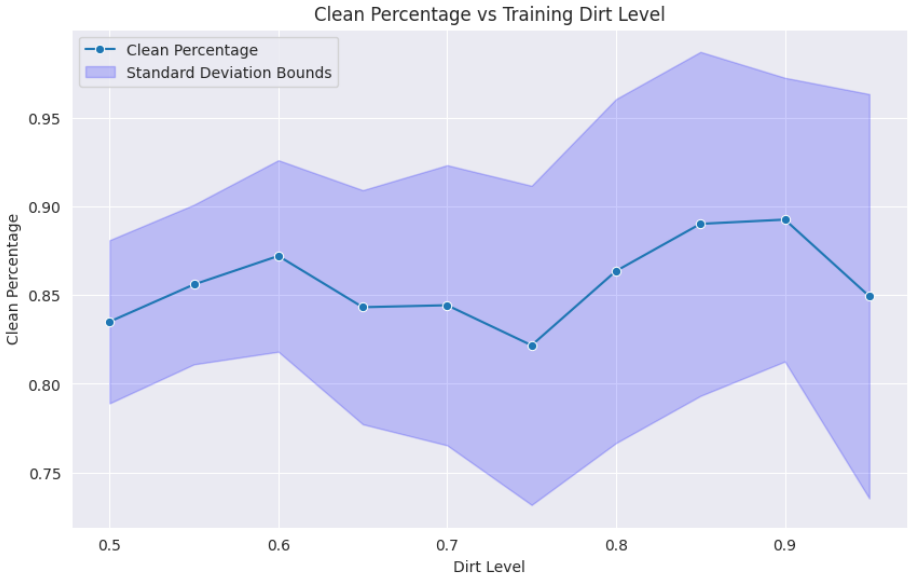


Figure 2: Cleaning effectiveness across different test environments (20%, 50%, 95% dirtiness) for various training dirtiness levels (0% - 100%). Include error bars representing standard deviation. Not shown is the random agent's performance at 19% average clean performance.

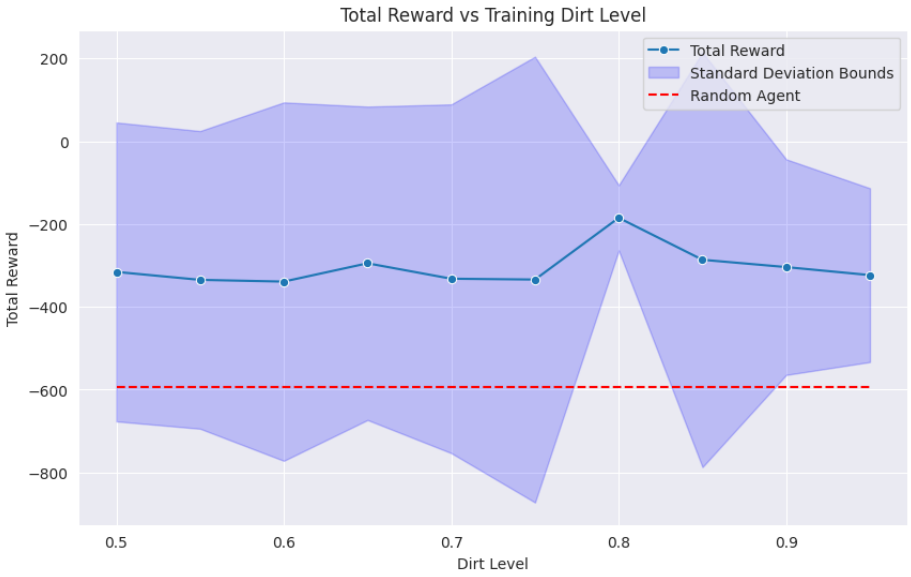


Figure 3: Total reward across different test environments (20%, 50%, 95% dirtiness) for various training dirtiness levels (0% - 100%). Include error bars representing standard deviation.

Table 1 summarizes the key statistical findings, providing p-values from two-tailed t-tests comparing the performance of agents trained at different dirtiness levels. These tests highlight the statistically significant differences in performance observed within and outside the optimal training dirtiness range (85%-95%).

Dirt level	0.55	0.6	0.65	0.7	0.75	0.8	0.85	0.9
Clean	0.856	0.8721	0.8432	0.8443	0.8217	0.8636	0.8902	0.8926
Clean Std.	0.045	0.054	0.066	0.079	0.09	0.097	0.097	0.08
Reward	-334.79	-338.982	-294.74	-332.06	-334.05	-184.89	-286.25	-303.93
Reward Std.	359.65	433.08	378.64	421.38	538.59	79.01	500.91	260.56

Table 1: Table of experiment statistics, clean and reward over a range of dirtiness levels and the standard deviation over these trials.

3.2 Impact of Battery State Granularity on Performance and Generalization

Next, we investigated the impact of the number of discretized battery states on agent performance. We trained agents with 2, 3, 4, and 5 battery states, each for 50,000 episodes in an environment with 85% static dirtiness. Generalization was then assessed across 20%, 50%, and 95% dirtiness test environments.

Table 2 presents the average cleaning effectiveness achieved for each battery state configuration across the different test environments. The table includes standard deviations to indicate the variability of results.

Table 2: Comparison of Cleanliness Scores Across Different Battery State Counts and Test Environments. * indicates $p < 0.05$, ** indicates $p < 0.01$.

Battery States	20% Dirtiness		
	Mean ± SD	Mean ± SD	p-value
5 vs. 4	0.8936 ± 0.1299	0.9355 ± 0.1449	0.037^{*}
5 vs. 3	0.8936 ± 0.1299	0.5738 ± 0.3476	$2e-15^{**}$
5 vs. 2	0.8936 ± 0.1299	0.2546 ± 0.1295	$2e-86^{**}$
4 vs. 3	0.9355 ± 0.1449	0.5738 ± 0.3476	$4e-18^{**}$
4 vs. 2	0.9355 ± 0.1449	0.2546 ± 0.1295	$8e-87^{**}$
3 vs. 2	0.5738 ± 0.3476	0.2546 ± 0.1295	$2e-15^{**}$

Battery States	50% Dirtiness		
	Mean ± SD	Mean ± SD	p-value
5 vs. 4	0.932 ± 0.1184	0.9599 ± 0.1051	0.08
5 vs. 3	0.932 ± 0.1184	0.6414 ± 0.3496	$2e-13^{**}$
5 vs. 2	0.932 ± 0.1184	0.4836 ± 0.2586	$8e-37^{**}$
4 vs. 3	0.9599 ± 0.1051	0.6414 ± 0.3496	$1e-15^{**}$
4 vs. 2	0.9599 ± 0.1051	0.4836 ± 0.2586	$9e-41^{**}$
3 vs. 2	0.6414 ± 0.3496	0.4836 ± 0.2586	$4e-4^{**}$

Battery States	95% Dirtiness		
	Mean ± SD	Mean ± SD	p-value
5 vs. 4	0.9367 ± 0.0532	0.9872 ± 0.0278	$8e-15^{**}$
5 vs. 3	0.9367 ± 0.0532	0.9268 ± 0.1147	0.43
5 vs. 2	0.9367 ± 0.0532	0.851 ± 0.1332	$1e-8^{**}$
4 vs. 3	0.9872 ± 0.0278	0.9268 ± 0.1147	$7e-7^{**}$
4 vs. 2	0.9872 ± 0.0278	0.851 ± 0.1332	$2e-19^{**}$
3 vs. 2	0.9268 ± 0.1147	0.851 ± 0.1332	$3e-5^{**}$

Figure 4 displays the average number of visits to each battery state during training for the different configurations. This visualization provides insights into the state utilization patterns learned by the agents and how these patterns relate to the number of available battery states.

In a 5-state battery the agent first enters in state 5. As the battery depletes, the agent enters in decrementing battery states. Likewise, an agent first enters state 4 in a 4-state battery, state 3 in a 3-state battery, and so on. State 1 is reserved for a dead battery.

Figure 5 is included to present a consistency in behavior that even lower battery state configuration counts display a learned reliance on the first entered battery state. The visitation count on state 3 increases as the visitation count on state 2 decreases, but still plays an active role at the end of training.

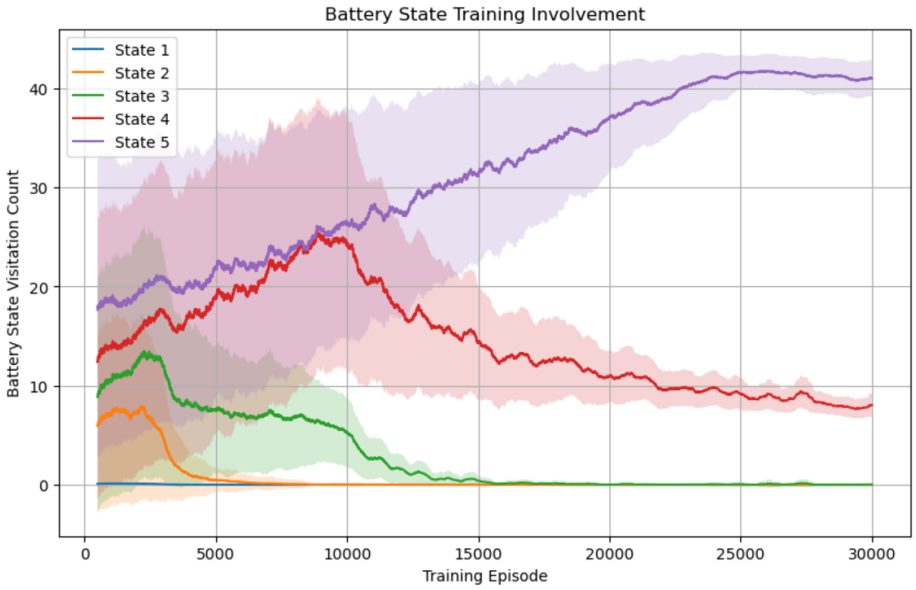


Figure 4: This graph shows the average number of environment steps spent in each battery state (shaded region 1 SD). It is a heuristic for how efficiently the agents use the vacuum and how converged the policy is. As training episodes advance, less battery is used, and the average battery use drops.

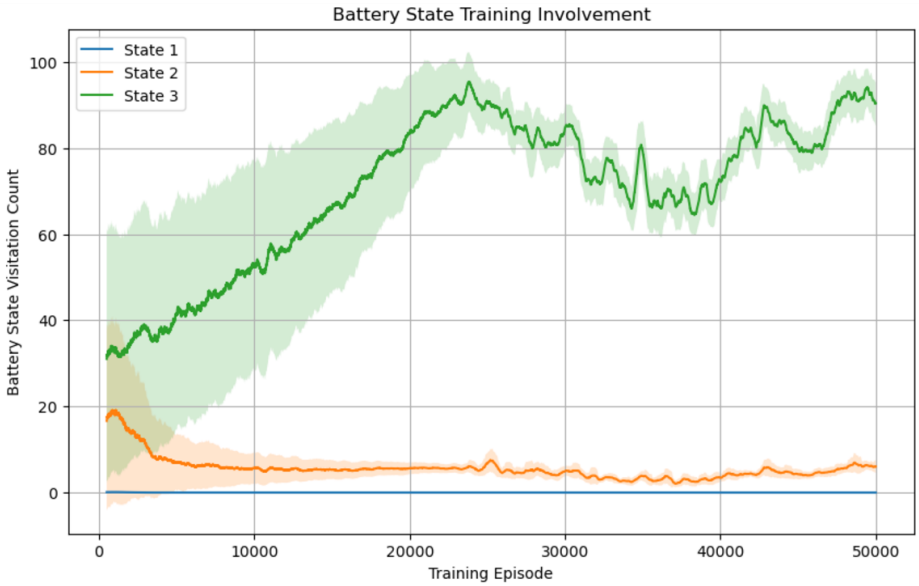


Figure 5: This graph shows the average number of environment steps spent in each battery state (shaded region 1 SD) with three battery states. Notice a similar trend of using fewer battery states as training continues.

4 Discussion

This section analyzes the implications of our findings, focusing on the impact of training dirtiness levels, battery state granularity, and generalization strategies, incorporating detailed observations from the experimental results.

4.1 Training Dirtiness Levels and Generalization

As detailed in Section 3, our experiments revealed a strong correlation between the training dirtiness level and the agent's generalization ability. Optimal generalization performance was observed within a narrow range of training dirtiness levels (approximately 85%-95%). This sensitivity to the training dirtiness level suggests that the learned policy is highly dependent on the distribution of dirtiness experienced during training. The sharp performance drop-off outside this optimal range underlines a lack of robustness and a limited ability to adapt to unseen environments. This behavior could stem from the reward structure, which incentivizes cleaning but does not explicitly incentivize efficient return-to-base strategies in low-dirtiness environments. Future research could explore alternative reward structures, curriculum learning, or meta-learning approaches to enhance generalization across a wider range of dirtiness levels.

4.2 Battery State Granularity and Performance: State Utilization and Generalization

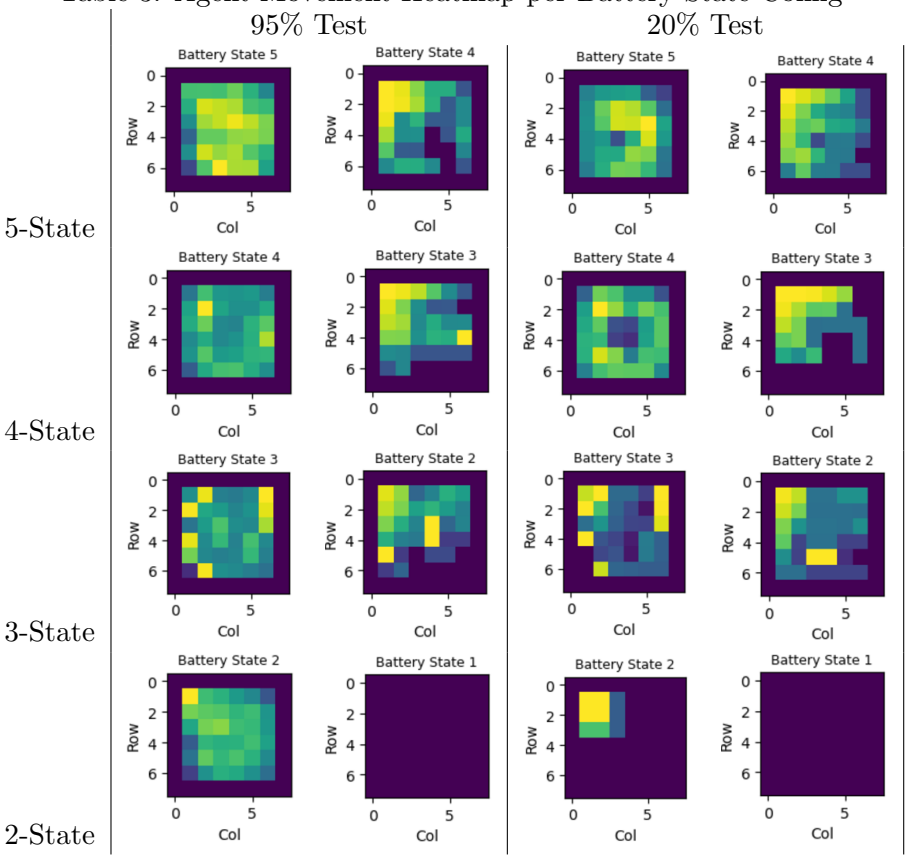
Our findings regarding battery state granularity underscore the critical role of state-space design in reinforcement learning. Section 3 shows that increasing the number of discretized battery states (from 2 to 5) significantly improved performance, particularly in terms of achieving high cleaning effectiveness across diverse test environments (20%, 50%, and 95% dirtiness). However, a fascinating observation is that as training progresses, agents tend to optimize their strategies, utilizing fewer battery states than initially available. Figure 4 vividly illustrates this phenomenon. For instance, the 5-state model, while initially exploring all states, converges to primarily using only states 4 and 5 after approximately 30,000 episodes. Likewise, figure 5 (3-state battery configuration), also shows the same behavior, although more erratically as training commences.

Despite this convergence to a smaller subset of states, increasing the initial number of battery states significantly boosts overall performance. This is evident in Table 2. The 4- and 5-state models consistently achieve over 89% cleanliness across all test environments, significantly outperforming the 2- and 3-state models. The high performance between all n-state models on the 95% dirtiness test compared to the significant difference between 4- and 5-state models to the lower state models on the 20% dirtiness test (p-value <0.001), suggests that the additional states in the 4- and 5-state models provides benefits mainly in scenarios deviating significantly from the training environment (85% dirtiness). This observation implies that a richer initial state space facilitates learning more effective and robust strategies, even if the agent eventually converges to using a smaller subset of those states. The increased computational cost associated with more states is justified by the significant gains in performance and generalization.

4.3 Spatial Exploration Patterns and Strategy Differences

The heatmaps (Table 3) provide compelling visual evidence supporting the hypothesis that a richer state space enhances the learning of robust and adaptable cleaning strategies. In the 5-state and 4-state models, the highest battery states, a full battery, exhibit efficient, widespread coverage across the testing environments. The next lower state, with a lower battery charge, demonstrates a concentration of activity near the home base (upper-left corner), clearly indicating a learned return-to-base strategy. This contrasts sharply with the 2- and 3-state models, which show considerably less systematic coverage, more sporadic activity, and instances of bright spotty tiles typically meaning the agent is inefficiently cycling back and forth.

Table 3: Agent Movement Heatmap per Battery State Config



The stark differences become especially evident in the 20% dirtiness test where there are more tiles as dark as the walls indicating a lack of visitation. For example, the 2-state agents primarily remained near the home base, achieving only approximately 25% coverage. Conversely, the 5-state agents maintained relatively high coverage, even in the 20% dirtiness environments. The 95% dirtiness tests illustrate this even further, as the 2-state agents are seemingly efficient (good coverage), yet significantly underperform with traversal in the lower dirtiness conditions. This suggests that the 2-state model may have converged to suboptimal strategies, demonstrating adequate average coverage but failing to

generalize to environments with lower dirtiness levels. These observations strongly support our hypothesis that a richer initial state space, while resulting in higher computational cost during training, enables the development of significantly more robust and effective cleaning strategies.

4.4 Limitations and Future Work

While our results demonstrate the significant benefits of a sufficiently granular battery state representation and appropriate training, our study has certain limitations. The use of tabular Q-learning, while providing interpretability, scales poorly to larger environments. Future research should explore alternative RL approaches, such as deep reinforcement learning and function approximation techniques, to address this limitation. The sensitivity of the model to the training dirtiness level also suggests the need to investigate advanced training methods like curriculum learning or meta-learning. Further research is warranted to investigate the interplay between reward design and the ability of an agent to achieve efficient return-to-base strategies while maintaining high cleaning effectiveness.

5 Conclusion

This research introduced the Robovac Problem, a novel reinforcement learning environment that realistically models the challenges of robotic vacuum cleaning with limited battery life and environmental uncertainty. Our experiments, using tabular Q-learning, systematically investigated the impact of two key factors—training dirtiness levels and battery state granularity—on agent performance and generalization.

Our findings demonstrate a critical relationship between training conditions and generalization ability. Optimal generalization was observed within a narrow range of training dirtiness levels (85-95%), highlighting the sensitivity of the learned policies to the training environment. Furthermore, increasing the number of discretized battery states significantly enhanced performance, particularly in terms of cleaning effectiveness and robustness across diverse test environments. This improvement is not simply due to increased exploration but to the learning of more nuanced and efficient cleaning and return-to-base strategies. Interestingly, agents converged to utilize a smaller subset of states during training, highlighting an ability to optimize their strategies. However, a richer initial state representation enabled the discovery of superior strategies.

The Robovac Problem provides a valuable benchmark for future research in resource-constrained RL, underscoring the importance of considering both training environment design and state-space representation for developing robust and generalizable agents. Future work should explore more sophisticated RL algorithms (e.g., deep reinforcement learning, function approximation), alternative reward shaping techniques, and advanced training methods like curriculum learning and meta-learning to further enhance agent performance and generalization capabilities in complex and uncertain environments. The nuanced insights gained from the Robovac Problem provide a valuable foundation for advancing research in resource-aware robotic autonomy.

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